INTRODUCTION TO NETWORK SYSTEMS · CLASS 5 (LIVE · IN THE ROOM)

Bits on a wave, and the whole stack

Modulation, multiplexing, the media close up and the answer to the 1200 Mbps question

Class 5 of 5 · the last class of the course

Dr. Thananop Thongthavorn
School of Information Technology, KMITL

THREE THINGS TODAY



The five questions

checked fast, then the gap the video left



Inside the plug

what a crimp does, and what breaks it



The whole course

Layer 7 to Layer 1 on one page

Phones out for the warm-up quiz. Pen and paper for everything else.

Where we are

The last one — and the only piece of the map still unshaded



done

Class 1

LIVE L2

Recap of L7–3, link layer, MAC, ARP



done

Class 2

VIDEO L2

Framing and error detection



done

Class 3

ONLINE L2

Multiple access, Ethernet, switches, VLANs, Wi-Fi



done

Class 4

VIDEO L1

Signals, line coding, media, bandwidth and capacity



today

Class 5

LIVE L1

Modulation, multiplexing, the media close up, whole-course wrap-up

Five classes, two layers, one stack. By four o’clock the whole thing closes.

WARM-UP · 5 MIN · FROM THE CLASS 4 VIDEO

Did you watch it?

Q1 A question says “a noiseless channel with 8 levels”. Which formula?

- A Shannon
- B Nyquist
- C either one
- D neither

Q2 An SNR of 20 dB is a power ratio of…

- A 2
- B 20
- C 100
- D 1000

Q3 Why can’t NRZ send a long run of 0s safely?

- A it uses too much power
- B the clock drifts
- C the CRC fails
- D it needs two wires

Q4 Doubling the bandwidth B of a channel roughly…

- A doubles capacity
- B adds one bit/symbol
- C does nothing
- D halves capacity

By the end of today, you can…

Six learning outcomes for the last class

1 — Block 1

Identify twisted pair, coax and fibre by sight, and say where each belongs

2 — Block 1

Explain what a crimped RJ45 does, and name two faults that break a link

3 — Block 2

Describe ASK, FSK, PSK and QAM, and read a constellation diagram

4 — Block 2

Compute a data rate from symbol rate and bits per symbol

5 — Block 3

Compare FDM, TDM, WDM and OFDM, and say where each is used

6 — Block 3

Give the complete answer to why you never see 1200 Mbps

Outcome 6 is the one I have been promising since Class 3. It needs all five classes to answer properly.

The layer you can touch

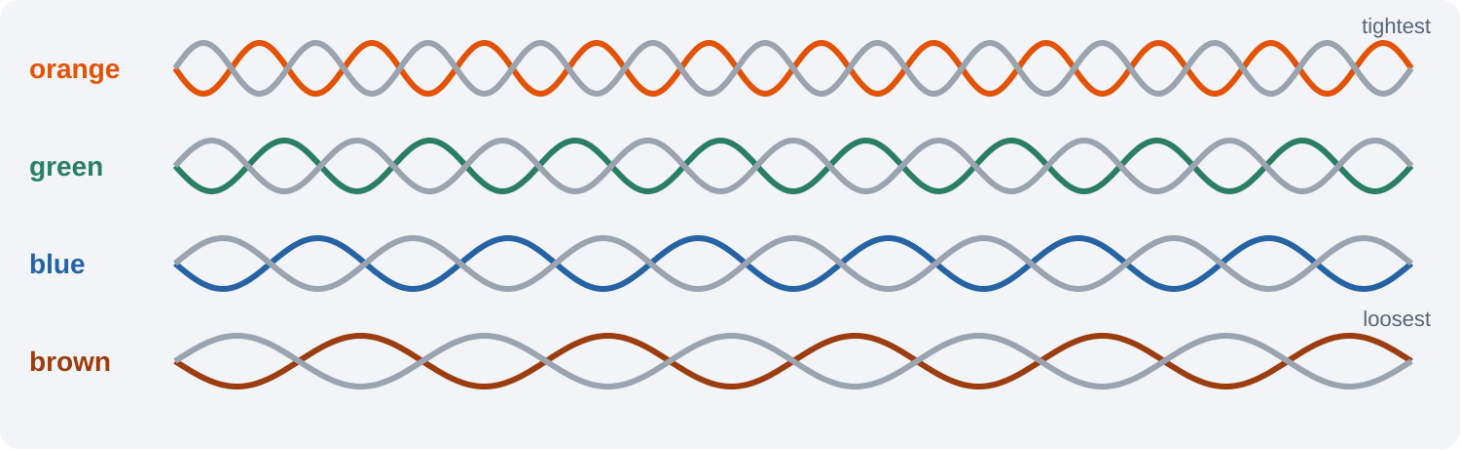


- The five cable questions from the video, checked fast
- Why green sits on pins 3 and 6 — the part the video skipped
- What happens inside an RJ45 plug when it is crimped
- What a cable tester actually tells you

STATION 1 · THE ANSWER

Every pair is twisted at a different rate

And that is deliberate, not sloppy manufacturing.



Why different rates?

If two pairs were twisted identically they would stay side by side along the whole cable, and each would keep inducing the same signal into the other. Different rates keep moving them past one another, so the crosstalk averages out.

The name for it

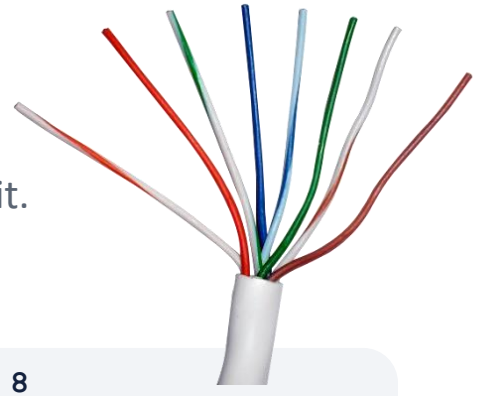
Signal leaking from one pair into its neighbour is crosstalk. It is the single biggest limit on how fast copper can go, and it is why Cat6 adds a plastic spine to hold the pairs apart.

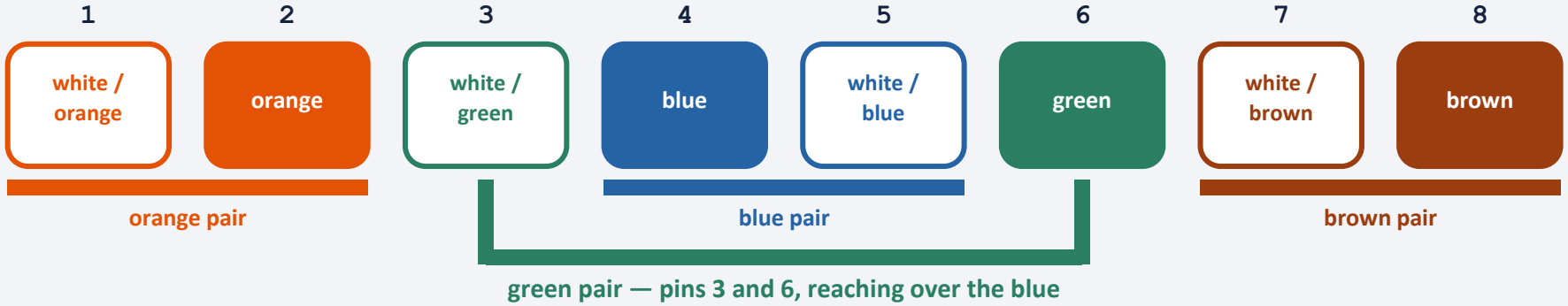
Back to Part 1 of the video: twisting makes external noise arrive equally on both wires so the receiver can subtract it. Varying the twist rate deals with the noise the cable makes against itself.

THE PART THE VIDEO SKIPPED

Why green sits on pins 3 and 6

Not 3 and 4, which would be neater — and the reason a miswired cable can pass every test and still fail at gigabit.





Why it was wired like that

Pins 4 and 5 in the middle belong to the blue pair — inherited from telephone wiring, where the centre pair carried the voice line. Green had no choice but to straddle it.

What happens if you ignore it

Put green on 3 and 4 and every wire still has continuity — the tester passes. But pins 3 and 6 now carry wires from two different twisted pairs. No cancellation: works at 100 Mbps, fails at gigabit.

What a cable tester actually tells you

Four faults, four signatures — this is the practical half of Layer 1



Open	One wire does not reach the contact, or is broken somewhere in the run.	*on the tester: one pin lights at the near end and never at the far end*
Short	Two conductors touching, usually from copper left outside the plug.	*on the tester: two pins light together, every time*
Reversed pair	The two wires of one pair are swapped.	*on the tester: the pair arrives, but in the wrong order*
Split pair	Continuity is perfect, but two wires come from different pairs.	*on the tester: every pin passes — and the link still fails at gigabit*

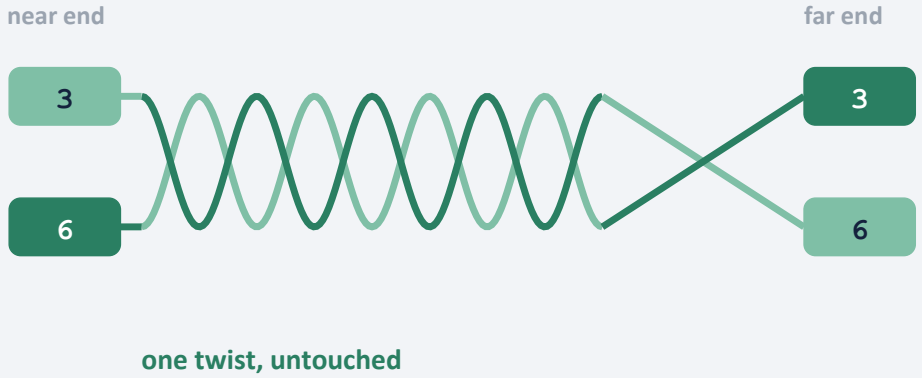
The split pair is the one worth remembering. A cheap tester passes it, it works at 100 Mbps, and it fails only at gigabit — when all four pairs finally carry signal.

TWO FAULTS THAT SOUND ALIKE

Reversed pair or split pair?

One gets the sign wrong and keeps the twist. The other keeps every pin in its place and destroys the twist.

REVERSED PAIR — the two wires of one pair, swapped

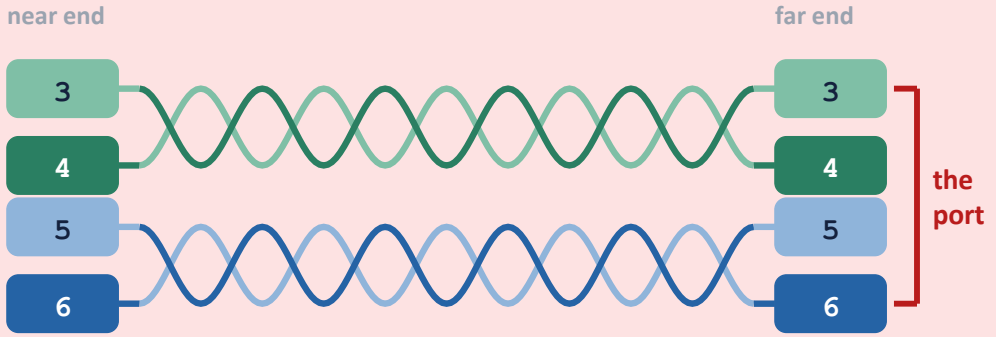


one twist, untouched

✗ wiremap FAILS — the tester shows it at once

The loop is unchanged — only the sign is. 100BASE-TX cannot even tell: MLT-3 carries data in transitions.

SPLIT PAIR — wires from two pairs used as one



two twists, one port straddling both

✓ wiremap PASSES — and that is the problem

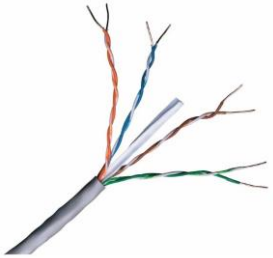
Every pin reaches its own number, so nothing looks wrong — but the port straddles two twists, and nothing cancels.

The tester reports the fault that does not matter, and misses the one that does

A wiremap tester compares pin numbers. A crossed pair is obvious to it; a split pair is invisible, because every pin does reach its own number. Finding a split pair needs a tester that measures crosstalk between pairs — which is most of what a certification tester is charging you for.

The category number on the jacket

Printed along the jacket every few metres — look at the next patch cord you plug in



Category	Bandwidth	Rated speed	Distance	What actually changed
Cat 5	100 MHz	**100 Mbps**	100 m	*obsolete — four pairs, only two used*
Cat 5e	100 MHz	**1 Gbps**	100 m	*the "e" is for enhanced: tighter crosstalk limits*
Cat 6	250 MHz	**1 Gbps / 10 Gbps**	100 m / 55 m	*adds a plastic spine to hold the pairs apart*
Cat 6a	500 MHz	**10 Gbps**	100 m	*the "a" is augmented — thicker, stiffer, better shielded*
Cat 7 / 8	600 MHz – 2 GHz	**10 – 40 Gbps**	100 m / 30 m	*individually shielded pairs; data-centre territory*

Notice the second column. The category is really a bandwidth rating in hertz — exactly the B from the video. Everything else follows from it.

And notice Cat 6 at 10 Gbps. It can do it — but only for 55 metres, not 100. Push the rate and the distance gives way. That is attenuation, in a product datasheet.

In practice almost every building is wired with Cat 6, and almost nobody needs Cat 7.

Why 100 metres, always

The one number every copper Ethernet standard refuses to change

Attenuation sets the ceiling

Copper loses signal with distance, and loses it faster at higher frequencies. Past about 100 m the signal at the far end is too close to the noise floor for the receiver to decide reliably.

And history fixed the number

The original 10BASE-T had to guarantee that a collision would reach the sender before it finished transmitting a 64-byte frame. That round trip, across repeaters and cable, works out at roughly 100 m per segment.

So the limit outlived its reason

Switched full-duplex Ethernet has no collisions at all, so the second reason has been irrelevant for twenty years. The first one is physics, and it has not moved. The number stayed.

A good exam sentence: “100 m is an attenuation limit that was originally also a collision-detection limit.” Both halves earn marks, and most answers give only the second.

One more thing that cable does

Power over Ethernet — because you have the pairs anyway

The idea

Put a DC voltage on the same pairs that carry data. The data is an alternating signal and the power is steady, so a simple filter at each end separates them — they do not interfere.

It is multiplexing, two blocks early: the medium is being shared between two completely different signals that never collide because they live in different parts of the spectrum.

Why it matters in practice

- a ceiling access point needs no electrician
- IP cameras, desk phones and door readers run on one cable
- the switch can reboot a frozen device by cutting its power remotely
- and it is why the IT department owns things that used to be the building's

`802.3af`

PoE · 15.4 W

phones, basic APs

`802.3at`

PoE+ · 30 W

pan-tilt cameras, better APs

`802.3bt`

PoE++ · 60 – 90 W

laptops, displays, LED lighting

Look at the ceiling in this room. Whatever is up there almost certainly has one cable going to it, not two.

Choosing a medium: four questions

The exam asks this as a scenario — these are the four things to check

1	**How far?**	Under 100 m, copper. Hundreds of metres, multimode fibre. Kilometres, single-mode. No cable possible, radio.
2	**How noisy is the environment?**	A factory floor, a lift shaft, anything near heavy motors: shielded copper, or fibre, which is immune to electrical interference entirely.
3	**Does it need to move?**	If the device moves, or you cannot run a cable to it, radio wins regardless of everything else — and you accept the rate and the sharing that come with it.
4	**What will it cost to change later?**	Fibre is expensive to install and cheap to upgrade; the glass outlives the equipment. Copper is the reverse.

The one that catches people: fibre is not “better” than copper. It is better at distance and immunity, and worse at cost per port for a desk twenty metres away. Answer the scenario, not the brochure.

BEFORE THE BREAK

You have now met every medium in this course.

In the video, Nyquist said: more signal levels means more bits per symbol.

So how does a radio build levels out of a wave in the first place?



10-minute break

Block 2 needs a calculator — have one with you when you come back.

Bits on a wave



- Why a signal needs a carrier at all
- Three things you can change: amplitude, frequency, phase
- QAM and the constellation diagram
- Why your Wi-Fi rate falls as you walk away

A QUESTION WORTH FIVE MINUTES

Why is an antenna a quarter of a wavelength?

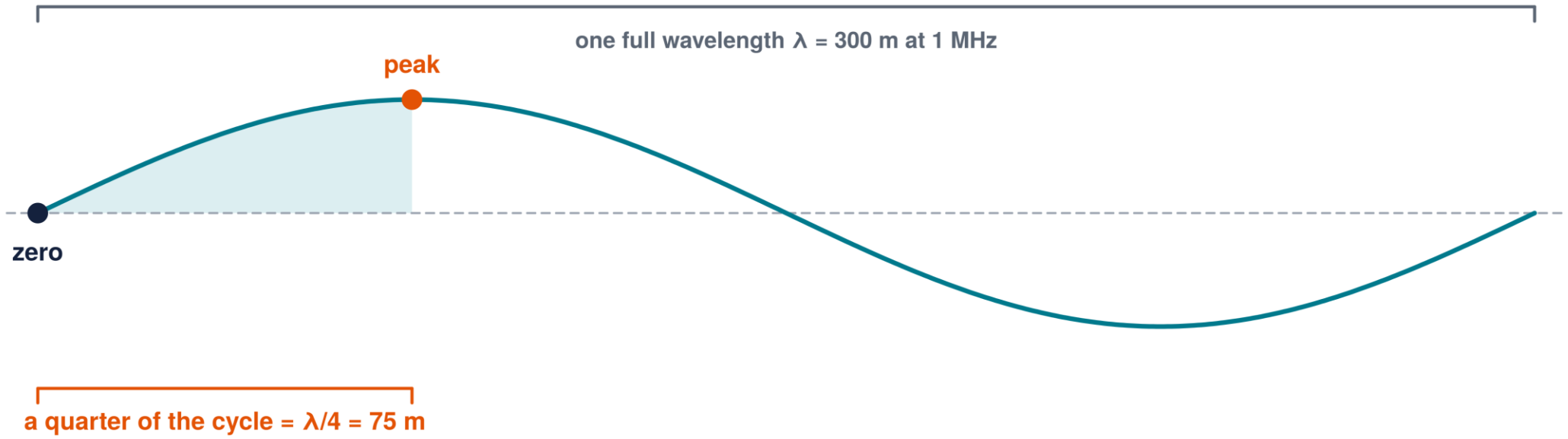
Two facts, and the answer falls out of them.

`λ = c / f  →  at 1 MHz, λ = 300 m  →  the antenna is 75 m`

STEP 1 · A FACT ABOUT SINE WAVES

A zero to the next peak is a quarter of a cycle

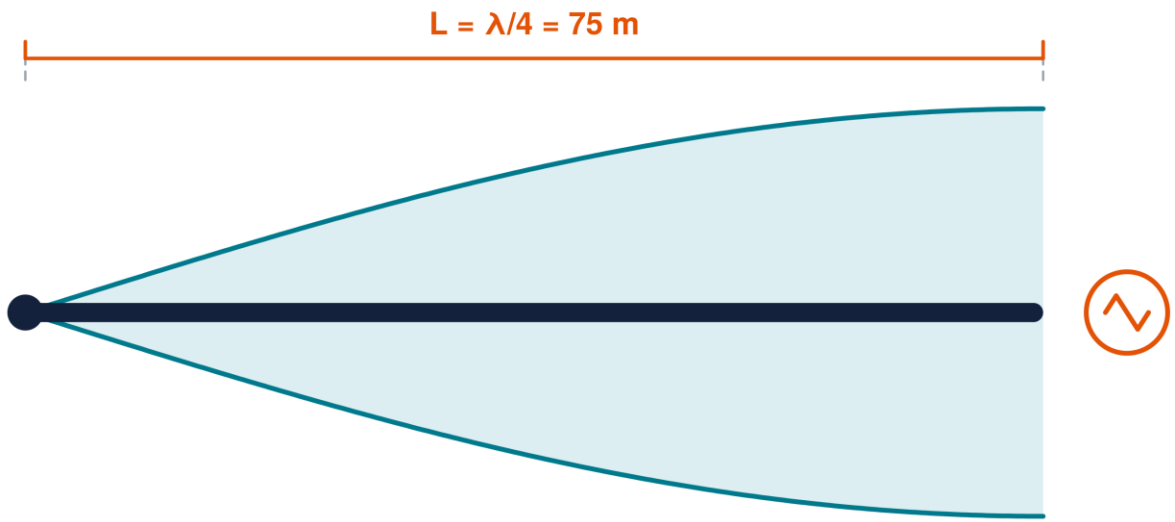
Nothing to do with antennas yet — this is just the shape of a sine.



STEP 2 · TWO THINGS THE WIRE CANNOT CHOOSE

The wire has to hold exactly that quarter

Current is what radiates — and its value is pinned at both ends of the antenna.



TIP

open end · current = 0

charge cannot flow into air

FEED

transmitter · current = maximum

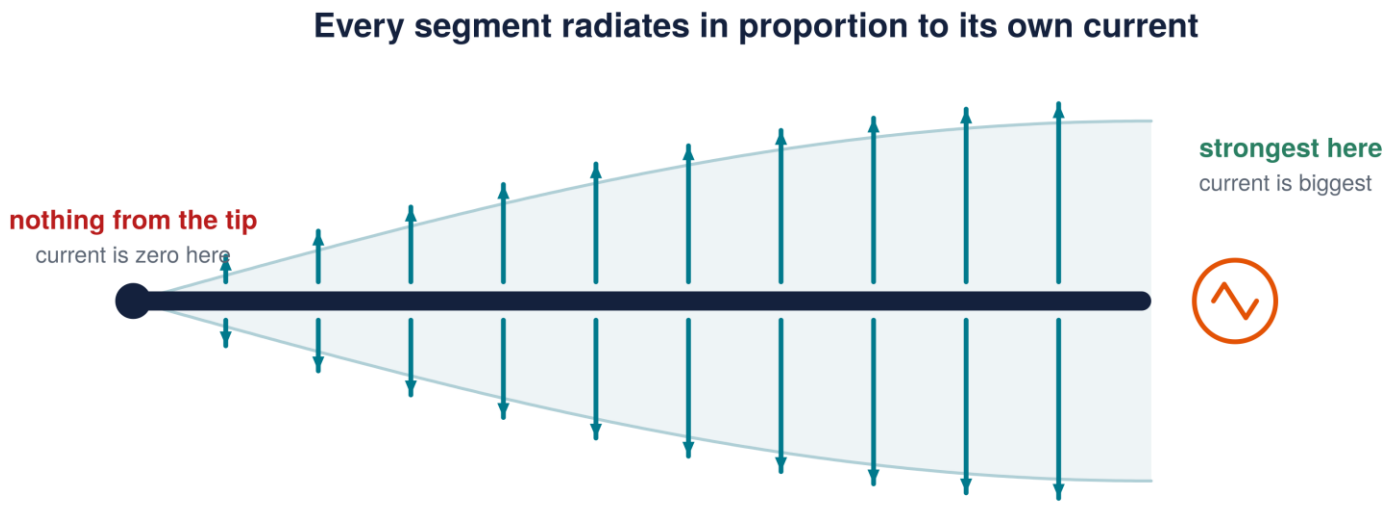
and current is what radiates

the curve is the current at each point along the wire

STEP 3 · WHERE THE WAVE ACTUALLY COMES FROM

The whole wire radiates — not the tip

There is no exit point. Every segment radiates in proportion to the current flowing in it.

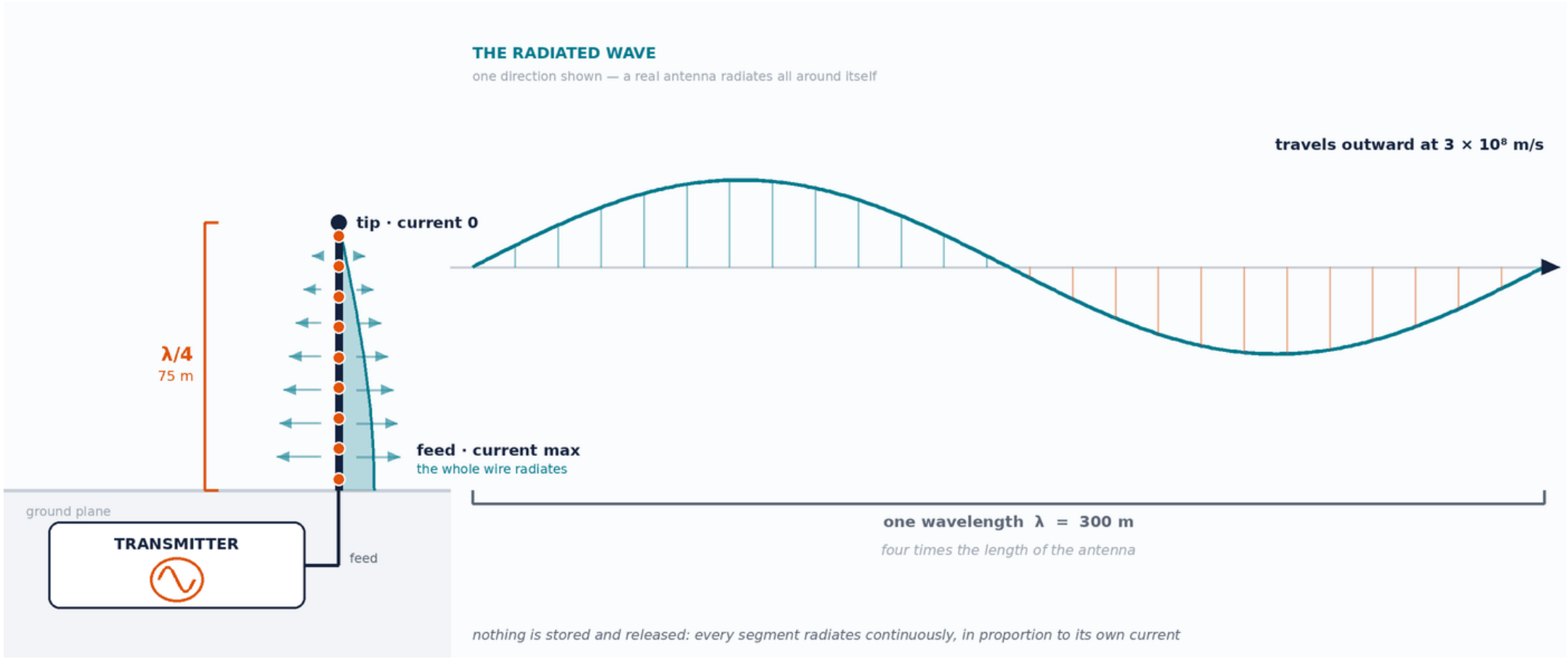


The tip is a boundary, not a nozzle: it forces the current to zero, which is what fixes the length.

PUT IT TOGETHER

The transmitter shakes the charge, the charge sheds the wave

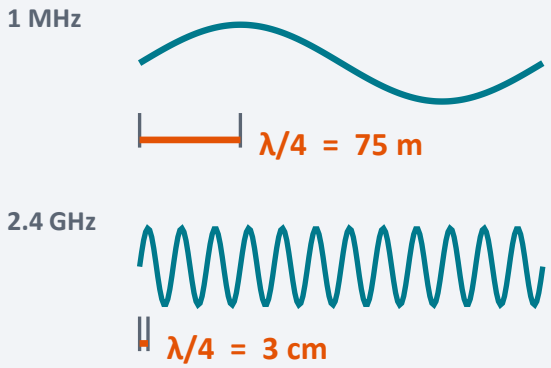
The wire oscillates in place. What leaves it travels outward at the speed of light.



Why not just send the square wave?

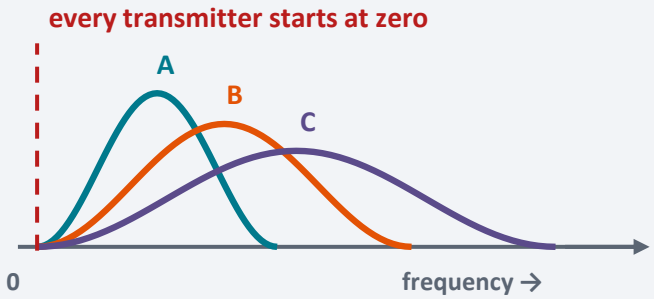
Baseband works on a wire. Through the air it does not — for three separate reasons.

1 An antenna must match the wavelength



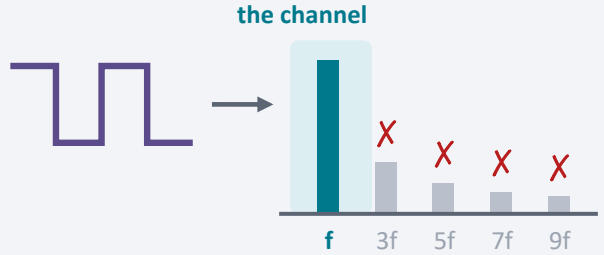
A useful antenna is about a quarter of a wavelength. At 1 MHz that is a 75-metre mast; at 2.4 GHz, 3 centimetres.

2 Everybody would sit on top of everybody



Baseband occupies the band from zero upwards, so three radios in one room occupy the same frequencies. There is nothing to tune to.

3 Square edges need bands the air will not give



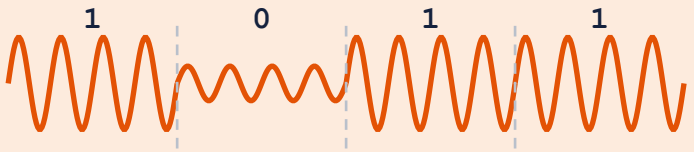
Only the fundamental gets through. Without its harmonics, the square collapses into a sine.



← what actually arrives

A square pulse is a fundamental plus a long tail of harmonics; a radio channel is a narrow slice.

So instead: take a steady sine at a frequency the channel likes — the carrier — and change something about it to carry the data. That is modulation, and it is the next hour.

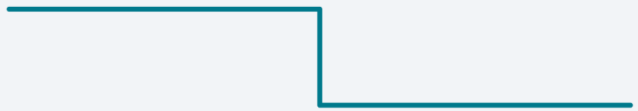


one carrier, its amplitude changed by the bits

“Nothing to tune to” — what it means

A signal is not AT one frequency. It occupies a RANGE of frequencies.

Why a data stream starts at zero



a long run of the same bit

the voltage barely moves → very low frequency, near 0 Hz



alternating as fast as possible

the fastest the stream can go → the top of the band, B

real data is a mix of both, so it fills

0 → B

BASEBAND — three radios transmitting as they are



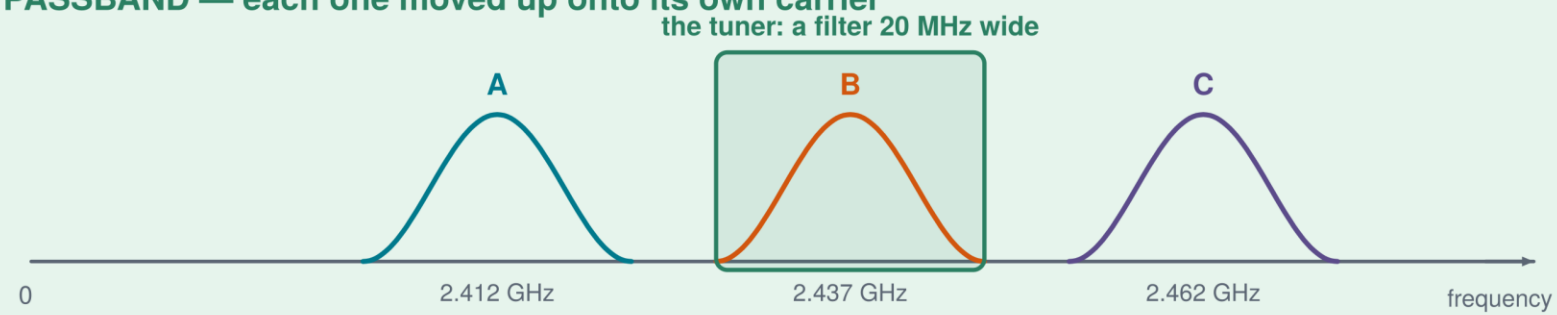
put your filter anywhere here

and you hear

A + B + C

all three start at zero — they sit on top of one another

PASSBAND — each one moved up onto its own carrier



tune to 2.437 and you hear

only B

A and C are filtered away

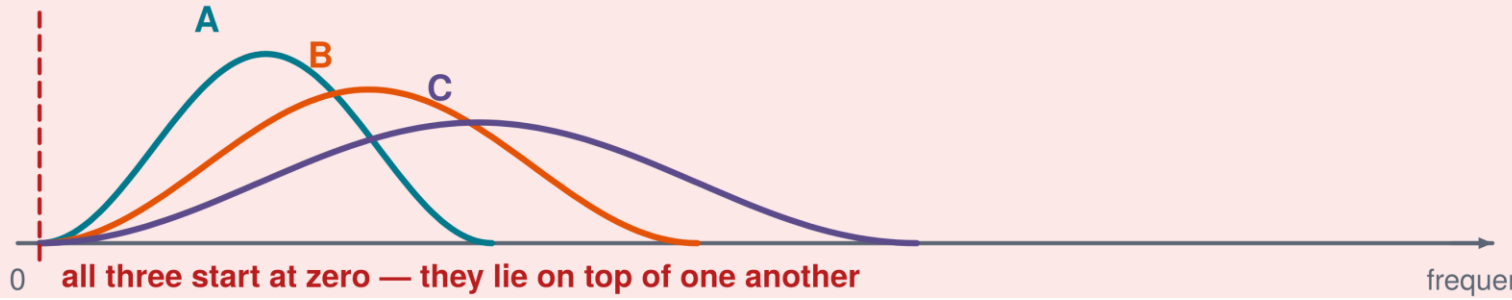
the carrier does not make the signal narrower — each is still 20 MHz wide. It moves it to its own address.

WHY A CARRIER AT ALL

“Nothing to tune to” — what it means

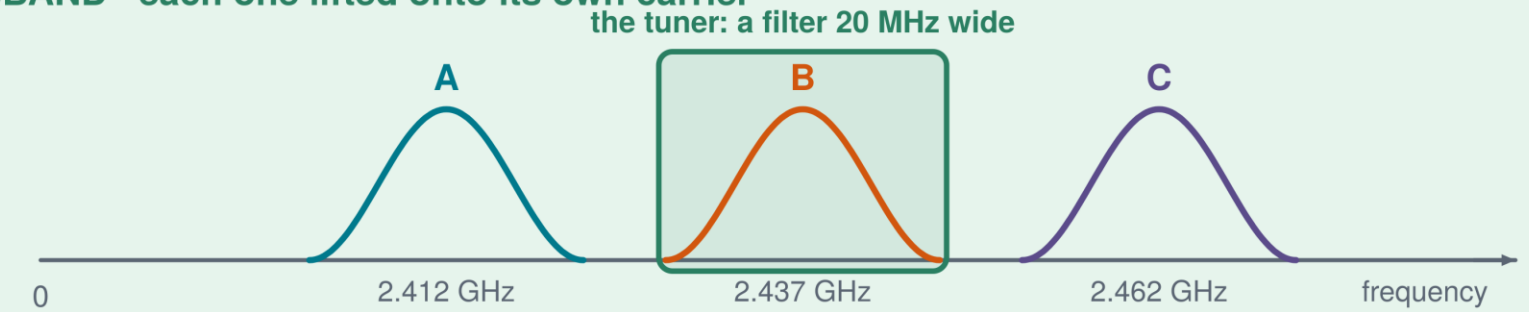
A signal is not at one frequency; it occupies a range. Both rows below hold the same three signals — only their position differs.

BASEBAND · three radios transmitting as they are



put your filter anywhere
and you hear
A + B + C

PASSBAND · each one lifted onto its own carrier

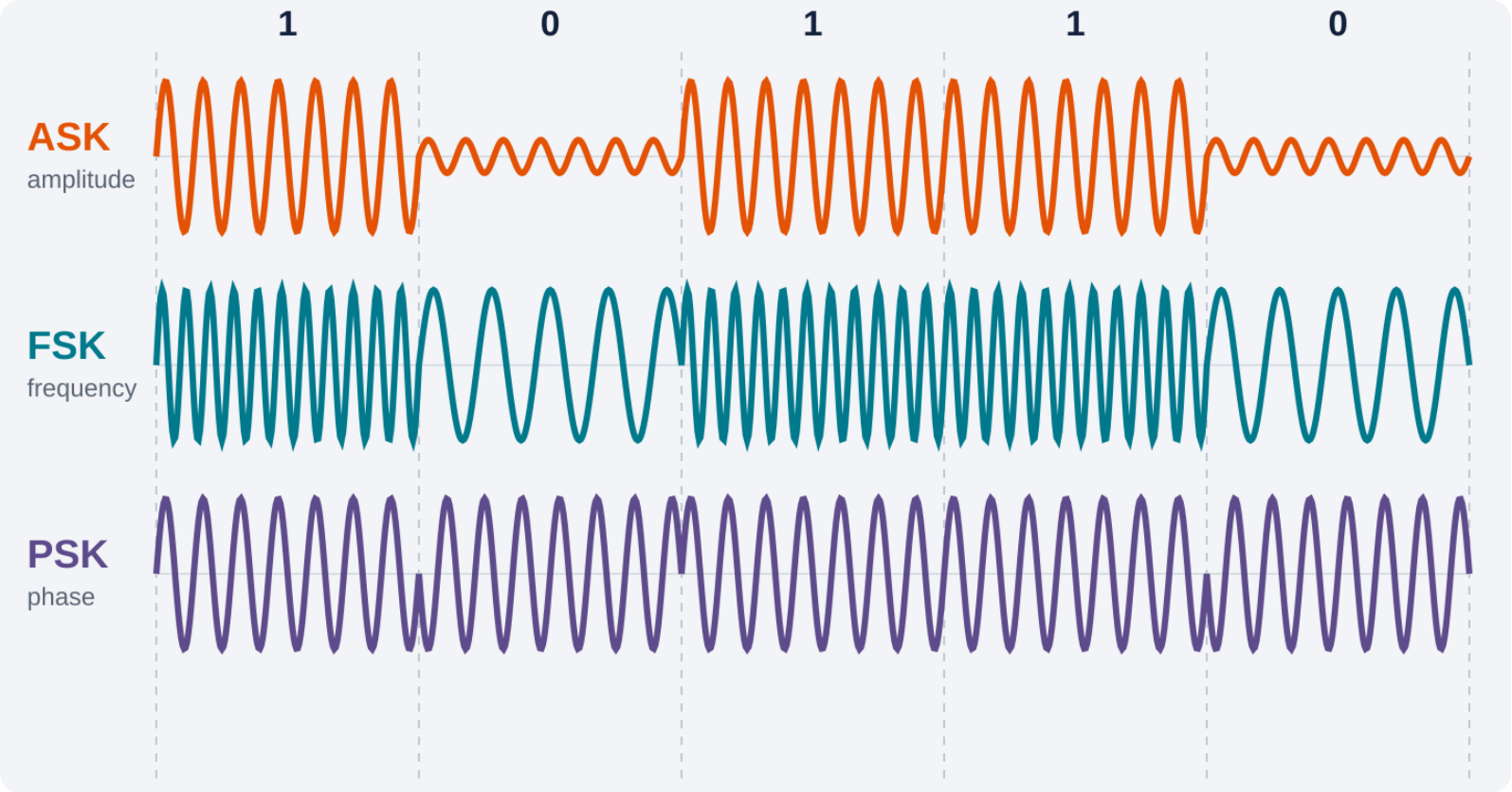


tune to 2.437 GHz
and you hear
only B
A and C are filtered away

Why “from zero upwards”: a long run of identical bits is a voltage that barely moves — a very low frequency. 1010 as fast as possible is the top of the band. Real data mixes both, so a baseband signal fills 0 → B. **And the carrier does not make a signal narrower — it moves it.** Each is still 20 MHz wide; it simply has an address.

A sine wave has exactly three things you can change

Each one gives a family of modulation schemes



Amplitude shift keying

Loud for a 1, quiet for a 0. Simple, and exactly what an optical fibre does — the light is on or off.

wrecked by noise, which is an amplitude change

Frequency shift keying

One tone for a 1, another for a 0. The old dial-up modem noise was this.

robust, but wasteful of bandwidth

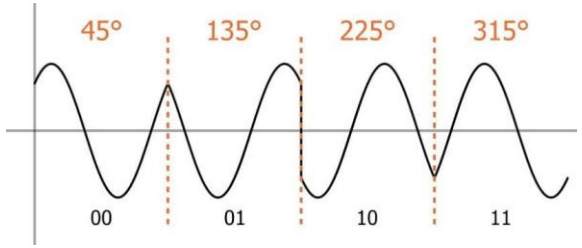
Phase shift keying

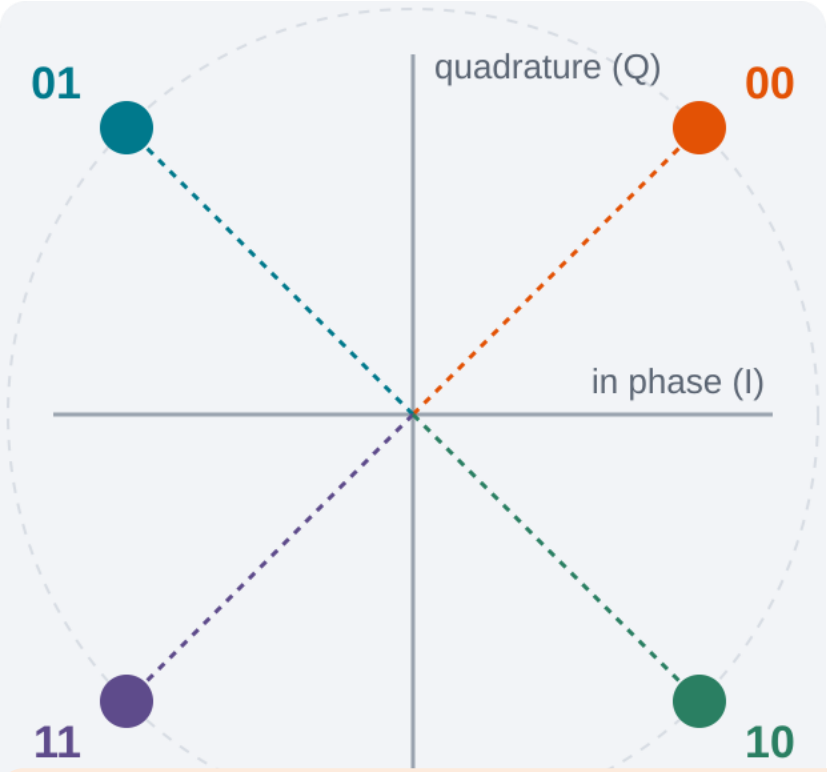
Same loudness, same pitch — the wave is flipped half a cycle. Hard to see, easy to measure.

the basis of everything modern

Four phases instead of two

The first step from one bit per symbol to two





What you are looking at: a constellation diagram

Every dot is one symbol the transmitter is allowed to send. How far the dot sits from the centre is the amplitude; the angle around the centre is the phase. Four dots means four possible symbols, and four symbols means two bits each — because $\log_2 4 = 2$.

The name

Four phases is QPSK — quadrature phase shift keying. It is what GPS, older Wi-Fi rates and a great deal of satellite still use, because it survives a poor signal and still carries two bits at a time.

Why stop at phase? Change amplitude as well, and the dots can sit anywhere on the plane, not just on a circle. That is the next slide, and it is where the big numbers come from.

QAM: use the whole plane

Quadrature amplitude modulation — amplitude and phase together



QPSK (4-QAM)

2 bits / symbol



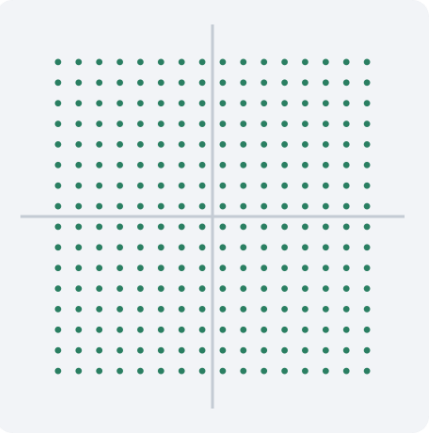
16-QAM

4 bits / symbol



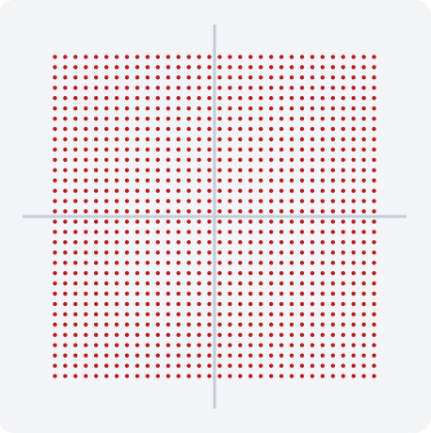
64-QAM

6 bits / symbol



256-QAM

8 bits / symbol



1024-QAM

10 bits / symbol

Every extra bit doubles the dots

M dots carry $\log_2 M$ bits. Going from 256 to 1024 adds only two bits per symbol — but it quadruples the number of levels the receiver has to tell apart.

And this is Nyquist's M

$C = 2B \cdot \log_2 M$ from the video was abstract. Here it is, physically: M is the number of dots in the picture, and the modulation scheme is how you build them.

Notice the dots getting closer together as you read left to right. Hold that thought for ninety seconds.

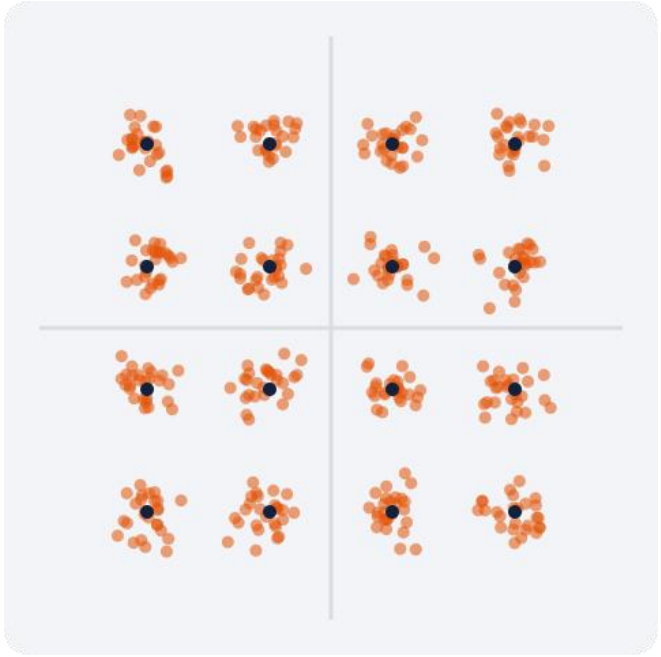
What noise does to a constellation

The same 16-QAM signal, received three times



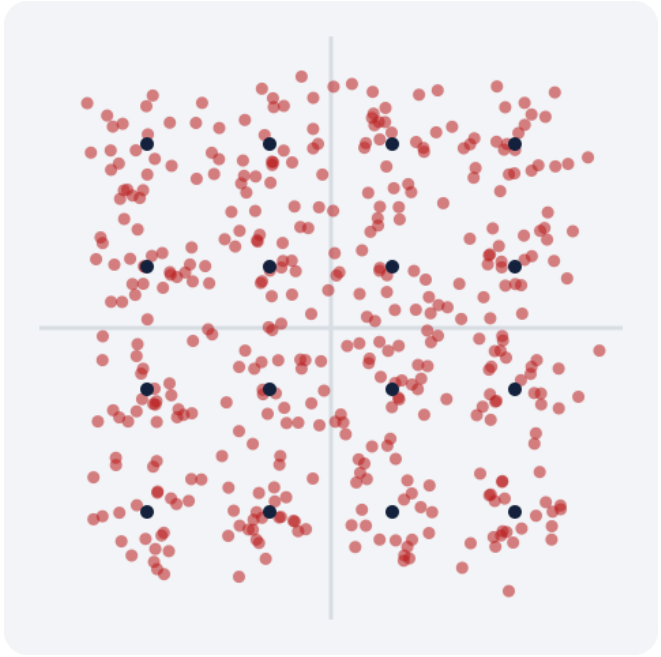
Strong signal

tight clusters — every dot lands unmistakably in its own box



Weaker signal

clouds spreading, but still separable by the receiver



Too far away

clouds overlap — the receiver guesses, and guesses wrong

This is Shannon, drawn. The noise cloud has a size set by the SNR. Pack the dots closer than the clouds are wide and symbols become indistinguishable — so the SNR decides how many dots you may use, which decides how many bits per symbol, which decides your rate.

So the radio changes modulation as you move

Adaptive modulation — the reason your Wi-Fi rate is never the number on the box

Modulation	bits / symbol	SNR it needs	Roughly where you are	Rate: 80 MHz, 2 streams
`1024-QAM`	10	~37 dB	*next to the router*	1201 Mbps
`256-QAM`	8	~31 dB	*same room*	961 Mbps
`64-QAM`	6	~21 dB	*next room*	576 Mbps
`16-QAM`	4	~13 dB	*down the corridor*	288 Mbps
`QPSK`	2	~8 dB	*two walls away*	144 Mbps
`BPSK`	1	~5 dB	*barely connected*	72 Mbps

Read the first and last rows together: walking from beside the router to two rooms away costs you a factor of sixteen, and nothing broke. The radio simply moved to a constellation with fewer dots, because the noise cloud grew.

SNR figures are typical receiver requirements and vary by chipset. Rates assume an 80 MHz 802.11ax channel, two spatial streams, 0.8 µs guard interval.

Working a data rate, end to end

Symbol rate × bits per symbol — and then the honest corrections

1	**Start with the symbol rate**	How many symbols the channel sends each second. Call it R_s.	***given, or 2B from Nyquist***
2	**Multiply by bits per symbol**	$\log_2 M$, read straight off the constellation.	***1024-QAM → 10 bits***
3	**Subtract the coding overhead**	Error-correcting codes add redundancy. A rate of 5/6 means five useful bits in every six sent.	***× 5/6***
4	**Multiply by the spatial streams**	Two antennas sending different data at once doubles it. This is MIMO.	***× 2***
5	**Now subtract everything from Class 3**	Headers, acknowledgements, backoff, beacons, and taking turns with everyone else.	***the big one***

Steps 1 to 4 give the number on the box. Step 5 is why you never see it. We put real figures through all five in Block 3.

Bit rate and baud rate are not the same number

A favourite exam trap, and a one-line defence against it

Baud rate (symbol rate)

How many times per second the signal changes to a new symbol. A property of the channel and its bandwidth — Nyquist capped it at 2B.

measured in baud, or symbols per second

Bit rate

How many bits arrive per second. Baud rate multiplied by the bits each symbol carries — which is a property of the modulation, not the channel.

measured in bits per second

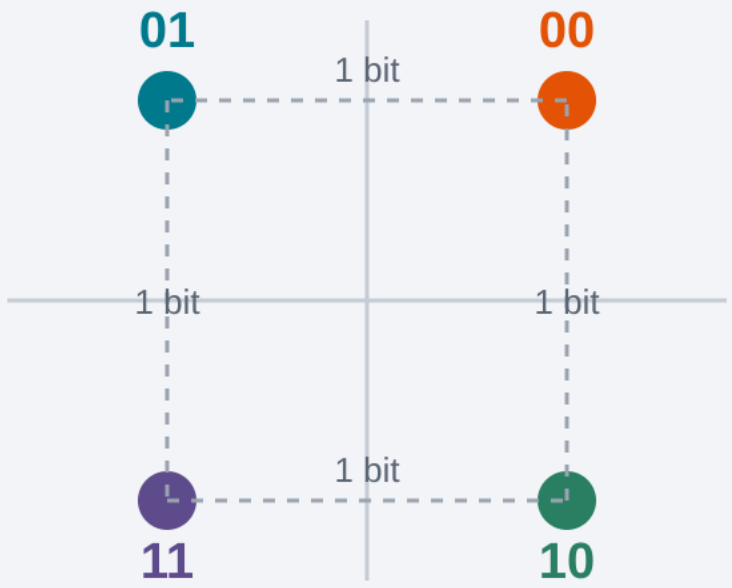
$$\text{bit rate} = \text{baud rate} \times \log_2 M$$

BPSK	QPSK	16-QAM	1024-QAM
1,000 baud × 1 bit	1,000 baud × 2 bits	1,000 baud × 4 bits	1,000 baud × 10 bits
`1,000 bit/s`	`2,000 bit/s`	`4,000 bit/s`	`10,000 bit/s`

Same channel, same baud rate, same bandwidth — ten times the bit rate. Only the constellation changed.

Why neighbouring dots differ by exactly one bit

Gray coding — a small idea that quietly halves your error rate



every neighbour: one bit different

The reasoning

When a symbol is decoded wrongly, it is almost never decoded as a dot on the far side of the diagram — noise nudges it into the dot next door. So if you arrange the labels such that every neighbouring pair differs in exactly one bit position, the overwhelming majority of symbol errors cost you one bit instead of two or three.

What it costs

Nothing. It is purely a choice of which bit pattern to assign to which dot. No extra power, no extra bandwidth, no extra bits — just a sensible numbering, and the bit error rate falls by roughly half.

Worth noticing as a pattern: some of the best engineering in this course costs nothing at all. Gray coding, twisting a pair of wires, putting the preamble at the front of a frame — all free, all enormous.

Spending bits to avoid resending frames

Forward error correction, and what a “5/6 coding rate” means

Class 2 detected errors and threw the frame away. Here we fix them where they land.

A CRC tells you a frame is damaged, and the only cure is a retransmission — which on a wireless link means waiting, contending for the channel again, and possibly failing again. Forward error correction instead sends extra bits alongside the data so the receiver can repair small amounts of damage on its own, immediately.

1/2

1 useful bit per 2 sent

100% overhead

a terrible signal, far from the AP

2/3

2 useful bits per 3 sent

50% overhead

a workable signal

3/4

3 useful bits per 4 sent

33% overhead

a good signal

5/6

5 useful bits per 6 sent

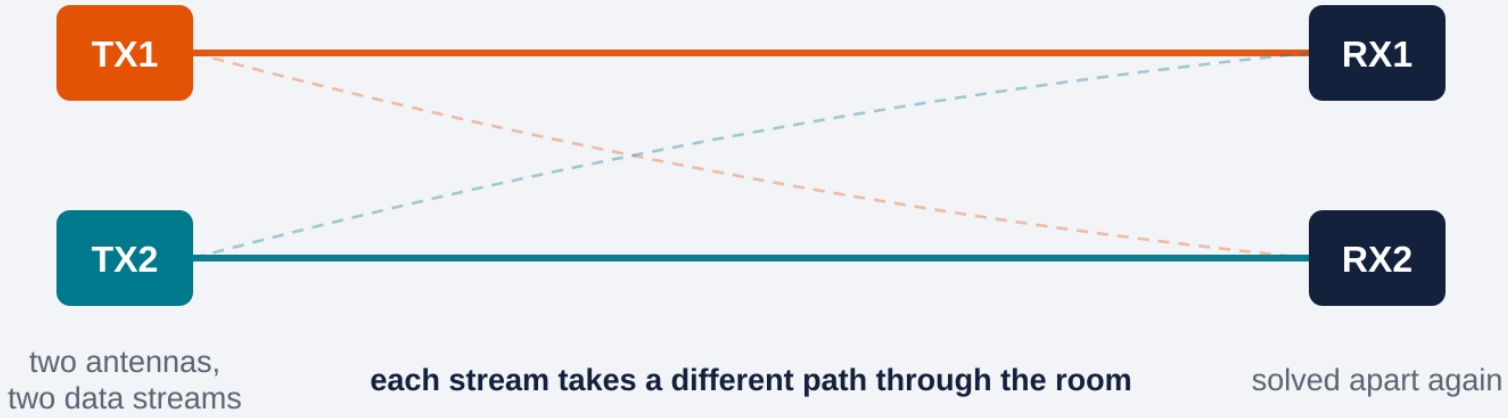
20% overhead

an excellent signal, close in

And the radio chooses this too, alongside the modulation. A weak signal gets fewer constellation dots AND a heavier code — so the rate falls twice over. That is why the drop as you walk away is steeper than the constellation picture alone suggests.

MIMO: the same channel, twice

Two antennas sending different data on the same frequency at the same time



How it is possible

Each receiving antenna hears a different mixture of the two streams, because each stream bounced off the room differently. Two measurements, two unknowns — the receiver solves the pair of equations and recovers both.

The surprise

Multipath — echoes off walls — used to be purely a problem. MIMO turns it into the resource that makes the trick work. A perfectly empty room is worse for MIMO than a furnished one.

And it is the only term in the whole calculation that beats Shannon's ceiling — because it is not one channel working harder. It is genuinely two channels, so the ceiling applies to each of them separately.

What everything around you actually uses

Same four ideas, different corners of the world

OFDM is not a modulation. It is many QAM signals side by side — Block 3.

Where	What it uses	Why that one
Optical fibre	`on / off keying (ASK)`	*glass is quiet, so the crudest scheme works and runs very fast*
Wi-Fi 6 (802.11ax)	`OFDM + up to 1024-QAM`	*hundreds of subcarriers, each with its own constellation*
4G and 5G	`OFDM + up to 256-QAM`	*the same machinery, with the operator scheduling who gets which subcarriers*
Gigabit Ethernet over copper	`PAM-5 on four pairs`	*five voltage levels, both directions at once, on all four pairs*
Digital TV and DAB radio	`OFDM + QAM`	*chosen for exactly the same multipath reason as Wi-Fi*
GPS	`BPSK with spreading codes`	*one bit per symbol — a satellite signal is weaker than the noise around it*

Read the last column down the page. Every choice is a trade between how quiet the medium is and how much complexity you can afford at each end. Fibre is quiet, so it is simple. Air is noisy and crowded, so it is elaborate.

YOUR TURN · 6 MIN · PAIRS

Three modulation problems

Calculator out. Write $\log_2 M$ as its own line each time.

1 **A channel sends 2 million symbols per second using 16-QAM. What is the raw bit rate?**

start with $\log_2 16$

2 **A link needs 48 Mbps. It can send 6 million symbols per second. How many dots must the constellation have?**

work backwards to M

3 **The same link in question 2 moves to a noisier room and drops to QPSK. What rate does it get now, and by what factor did it fall?**

two answers

Ignore coding and streams for now. Raw symbol rate times bits per symbol, nothing else.

ANSWERS

Check your working

Question 3 is the one that matters — it is adaptive modulation in a single number.

1 2 Msym/s, 16-QAM

$\log_2 16 = 4$ bits per symbol

$2{,}000{,}000 \times 4$

= 8 Mbit/s

2 48 Mbps from 6 Msym/s

$48{,}000{,}000 \div 6{,}000{,}000 = 8$ bits

$M = 2^8$

= 256 dots, so 256-QAM

3 The same link, now QPSK

QPSK = 2 bits per symbol

$6{,}000{,}000 \times 2 = 12$ Mbit/s

a quarter of the rate

Question 3 in words: nothing about the link broke. The cable did not fray, the router did not fail, the bandwidth did not change. The SNR fell, so the receiver could no longer separate 256 clouds, and the transmitter fell back to 4. Everything else followed.

That single mechanism explains most of the gap between an advertised speed and a measured one.

QUICK CHECK · 3 MIN

Modulation

Q1 A constellation with 64 dots carries…

- A 6 bits/symbol
- B 8 bits/symbol
- C 64 bits/symbol
- D 32 bits/symbol

Q2 Which modulation is most vulnerable to noise?

- A FSK
- B PSK
- C ASK
- D all the same

Q3 Your Wi-Fi rate falls as you walk away because…

- A the router slows down
- B fewer constellation dots are usable
- C the cable is longer
- D the CRC fails

Q4 On a constellation, the angle around the centre is the…

- A amplitude
- B frequency
- C phase
- D bit rate

BEFORE THE BREAK

One carrier, one conversation. That is a waste of a fibre.

A single strand of glass can carry roughly four thousand gigahertz of spectrum.

So how do you put hundreds of separate conversations on it at once?



10-minute break

After it: multiplexing, the 1200 Mbps answer, and the whole course on one page.

Sharing, and summing up



- Multiplexing: FDM, TDM, WDM and OFDM
- The complete answer to the 1200 Mbps question
- Layer 7 down to Layer 1 on one page
- What the exam actually asks

You have met this problem before

Class 3, Block 1 — only the answer is different now

In Class 3 we asked:

many devices, one channel — who is allowed to talk?

The answer was a protocol: take turns, or listen first, or allow collisions and recover. Nobody is sending at the same instant as anybody else.

Today we ask:

one medium, many signals — how do they all fit at once?

The answer is physical: give each signal its own slice of frequency, or time, or colour, so they genuinely travel together and never interfere.

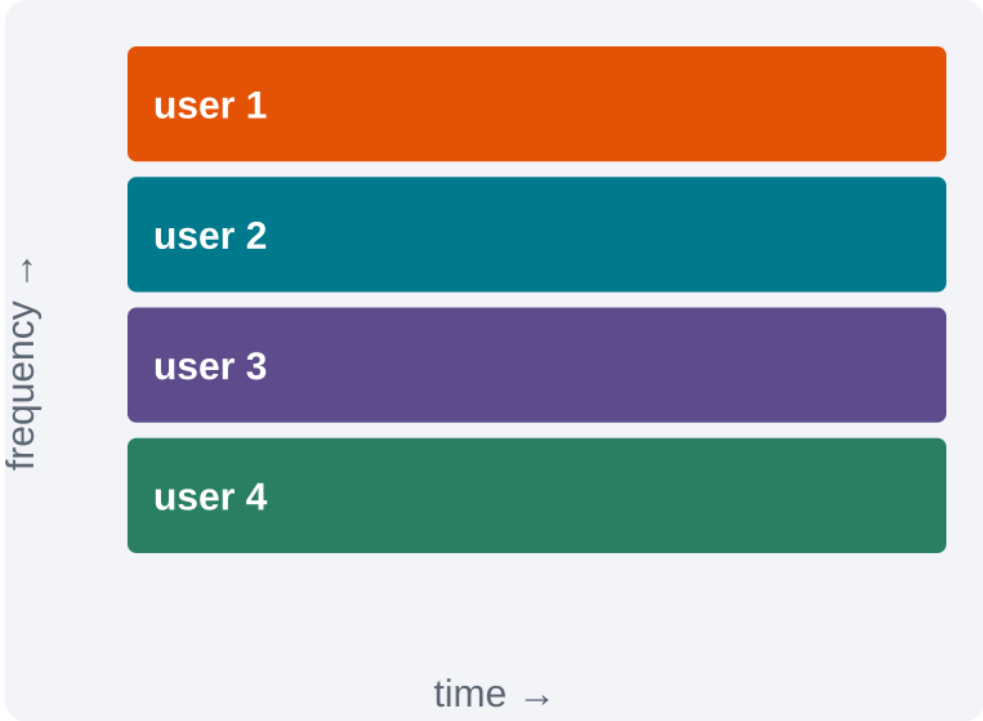
Multiple access and multiplexing are the same mathematics, used by different people

Channel partitioning in Class 3 — TDM and FDM — is exactly multiplexing. The difference is who is doing it and why. Multiple access shares one channel between competing devices that cannot see each other. Multiplexing shares one medium between signals that one operator controls, usually to avoid laying a second cable.

So if the Class 3 TDM and FDM slides felt familiar — they were a preview of this.

The two originals: split the frequency, or split the time

Same medium, two different ways to carve it up



FDM — frequency division

Each user keeps one band, all of the time. Radio and TV broadcasting, the old analogue telephone trunks, and every Wi-Fi channel number you have ever picked.



TDM — time division

Each user gets the whole band, for a slot at a time, forever repeating. Digital telephony, T1 and E1 lines, and the backbone of mobile voice.

Class 3 called these channel partitioning

same maths, different job

TDM in practice, and its one weakness

The telephone network ran on this for fifty years

Synchronous TDM

Slot 3 belongs to user 3, every frame, whether or not user 3 has anything to say. Simple, perfectly predictable, and the delay never varies — which is exactly what a voice call needs.

T1 (North America, Japan)	`24 channels × 64 kbps + 8 kbps framing = 1.544 Mbps`
E1 (Europe, Thailand)	`32 channels × 64 kbps = 2.048 Mbps`

Statistical TDM

Give each slot to whoever actually has data waiting. Idle users get nothing, so the medium is never wasted — but now a slot needs a label saying whose it is, and the delay varies with how busy things are.

A label on every slot saying who it belongs to, and variable delay, is a packet network. You have been studying one all term.

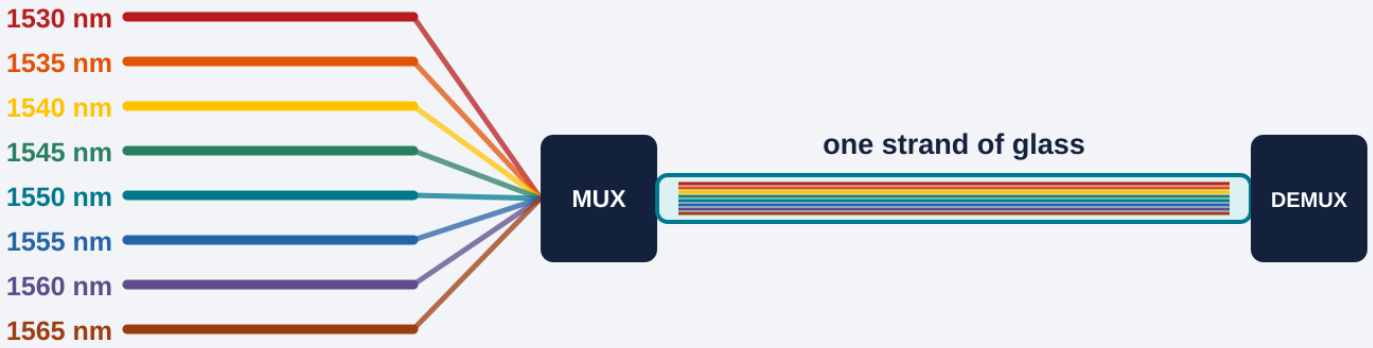
The weakness, and why packets won

Synchronous TDM wastes a slot every time its owner is silent. On a voice call that is fine — people talk steadily. On data traffic, which is bursty, it is catastrophic: most slots carry nothing while other users queue. That single observation is why the telephone companies eventually rebuilt their networks around IP.

Compare with the Class 3 verdict on TDM: a single active node still gets only one slot in six. Same objection, bigger network.

WDM: frequency division, done with colour

The reason a single fibre can carry a continent



It is just FDM

Each signal gets its own wavelength — its own colour — and a prism-like device combines them at one end and separates them at the other. The glass carries all of them at once without knowing it.

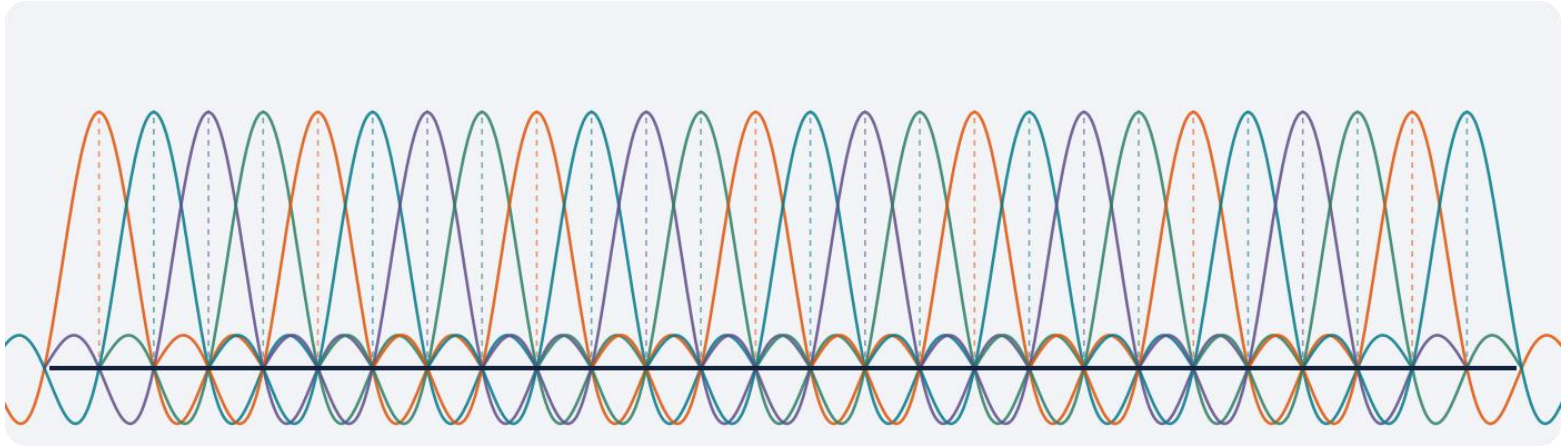
The numbers

- the C-band alone is about 4.4 THz wide
- at 50 GHz spacing that is ~88 channels
- at 100 Gbps each: ~8.8 Tbps on one fibre

And this is why laying fibre is worth it. To increase capacity you do not dig the road up again — you change the equipment at each end and light more colours in glass that is already buried.

OFDM: the one your phone is using right now

Hundreds of narrow carriers instead of one wide one



frequency → each peak sits exactly on its neighbours' zeros

Orthogonal

The carriers overlap, yet each one peaks exactly where every other is zero. So they can be packed far tighter than FDM would allow, with no guard bands wasted between them.

Why bother

A narrow carrier has a long symbol, so echoes off walls arrive well inside one symbol instead of smearing into the next. It is the fix for multipath.

Where it is used

Wi-Fi, 4G, 5G, DSL and digital TV. Essentially all modern wireless.

And each subcarrier is separately modulated — its own little constellation. An 80 MHz Wi-Fi channel (802.11 ac) has 980 of them carrying data, and the radio can use a different constellation on each if part of the band is noisy.

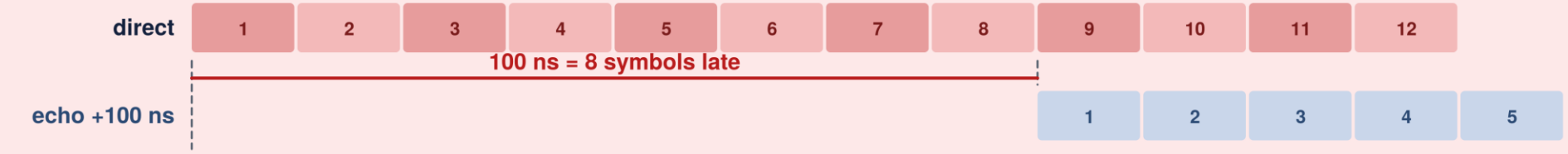
THE REASON FOR ALL THOSE SUBCARRIERS

Long symbols are what beat the echo

An echo is a copy that went further: 100 ns late means 30 m of extra path.

ONE WIDE CARRIER · 80 MHz, so each symbol lasts 12.5 ns

ruler below: 150 ns across



symbol 1 of the echo lands on symbol 9 of the direct signal — every symbol is corrupted by its predecessors

ONE OFDM SUBCARRIER · 78.125 kHz, so each symbol lasts 12.8 µs

ruler below: 30 µs across — the same picture, zoomed out 200×



the echo never reaches the next symbol, and the receiver throws the guard away before reading the symbol

So why not simply slow one carrier down? Because symbol length and data rate fight each other — one carrier with 12.8 µs symbols sends 78,000 symbols a second. OFDM buys long symbols without paying for them: **a thousand slow subcarriers at once — slow each, fast together.**

Four ways to share one medium

Learn the first column and the last — the exam asks for both

Scheme	Divides the medium by	How it works	Where you meet it
FDM	**frequency**	Each signal keeps its own band permanently	*Radio and TV broadcast, Wi-Fi channel numbers, analogue phone trunks*
TDM	**time**	Each signal gets the whole band in repeating slots	*T1 and E1 lines, digital telephony, mobile voice*
WDM	**wavelength**	FDM on fibre: one colour of light per signal	*Every long-distance and undersea fibre link*
OFDM	**many narrow carriers at once**	Hundreds of orthogonal subcarriers, each separately modulated	*Wi-Fi, 4G, 5G, DSL, digital television*
CDM	**code**	All signals share all the time and all the band, separated by orthogonal codes	*3G mobile, GPS*

One sentence that holds all of it: a medium has frequency, time and — in fibre — colour. Any of those can be cut into pieces and handed out, and every scheme in the table is one of those three choices, or a mixture.

YOUR TURN · 5 MIN · PAIRS

Four scenarios — which scheme?

Name the multiplexing scheme, and say in one clause why that one.

1. **A single buried fibre between Bangkok and Chiang Mai must carry 40 separate 100 Gbps links.**

2. **A 2 MHz band must be split permanently between four radio stations that broadcast continuously.**

3. **An indoor wireless link suffers badly from echoes off walls and furniture.**

4. **Thirty-two voice calls must share one 2.048 Mbps line with a fixed, predictable delay.**

One of these has two defensible answers. Decide which, and be ready to argue for it — that is where the marks are in a scenario question.

ANSWERS

Four scenarios

Question 2 is the one with room to argue.

1	**WDM**	One fibre, many signals, and the only thing you can divide in glass is wavelength. 40 channels is well inside what the C-band holds.
2	**FDM**	A permanent band each, broadcasting continuously, so there is nothing to gain from time slots. This is radio broadcasting as it has always worked.
3	**OFDM**	Echoes are multipath. Narrow subcarriers mean long symbols, so the echo lands inside the same symbol instead of smearing into the next.
4	**TDM — synchronous**	Fixed, predictable delay rules out statistical TDM. And 32 × 64 kbps = 2.048 Mbps exactly: this is an E1 line.

Question 2 could also be OFDM, and digital radio genuinely does use it. If you said that and explained why, it is a correct answer — the examiner is reading your reason, not matching a word.

The five classes, in one line each

If you can expand each line into a paragraph, you are ready

1	**The stack, and finding your neighbour**	A request from a browser travels down seven layers. Layer 2 needs a MAC address, and ARP is how a device finds one. Local delivery versus global delivery.
2	**Finding the frame, and trusting it**	Byte and bit stuffing, the preamble and SFD, parity, the Internet checksum, and CRC — which catches every burst up to the length of its polynomial.
3	**Who is allowed to send**	Partition, contend, or take turns. ALOHA through CSMA/CD, the 64-byte minimum, self-learning switches, broadcast domains, VLANs, and why Wi-Fi had to be different.
4	**How a bit becomes physical**	Two delays, three impairments, decibels, the clock problem, line codes from NRZ to 64b/66b, the media, and the Nyquist and Shannon ceilings.
5	**How a bit rides a wave, and how many fit**	Modulation and constellations, adaptive rate, multiplexing from FDM to OFDM — and the full account of the gap between a specification and a speed test.

Notice how the lines get more physical as you read down. That was deliberate: we started with what the network promises and ended with what the universe permits.

Six things people get wrong in the exam

Every one of these has cost real marks

✕ “Bandwidth is speed”

Bandwidth is hertz; data rate is bits per second. They are related by the modulation, not equal.

✕ Putting dB straight into Shannon

Convert first: S/N = 10^(dB/10). Thirty decibels is a ratio of one thousand, not thirty.

✕ “A switch splits broadcast domains”

It splits collision domains. Only a router splits broadcast domains.

✕ “A faster link reduces all delay”

It shortens L/R only. Propagation delay d/s does not care how fast your card is.

✕ “CRC fixes errors”

CRC detects. Fixing is either a retransmission or a separate error-correcting code.

✕ “Fibre is always better”

Better at distance and immunity. Worse at cost per port for a desk twenty metres away.

Read this slide again the night before. Avoiding these six is worth more than learning anything new at that point.

QUICK CHECK · 3 MIN

Multiplexing

Q1 Which scheme divides a fibre by colour?

- A FDM
- B TDM
- C WDM
- D CDM

Q2 OFDM resists multipath because its subcarriers are…

- A wide, so symbols are short
- B narrow, so symbols are long
- C louder
- D fewer

Q3 A slot wasted when its owner is silent is a feature of…

- A statistical TDM
- B synchronous TDM
- C OFDM
- D WDM

Q4 Class 3 channel partitioning and multiplexing are…

- A unrelated
- B the same mathematics
- C opposites
- D both Layer 3

Where 1200 Mbps comes from — and where it goes

The question from Class 3, answered in full

in practice: 300 – 600 Mbps on a good day

Building the number on the box

`980 data subcarriers`	*an 80 MHz OFDM channel*
`× 10 bits per symbol`	*1024-QAM, 32 × 32 dots*
`× 5/6`	*the error-correcting code*
`÷ 13.6 µs`	*one symbol plus its guard interval*
`× 2`	*two spatial streams — MIMO*

= 1,201 Mbps

Why you measure a fraction of it

The signal is never perfect	so the radio drops to a smaller constellation	today
Headers on every frame	Ethernet, IP and TCP all take their cut	C1–2
An ACK for every frame	and the air is idle while you wait for it	C3
Backoff, beacons, contention	the channel is shared, and taking turns costs time	C3
Half duplex: never both ways at once	your download and its acknowledgements compete	C3

Neither side is a lie. The left column is a correct calculation about the radio. The right column is everything the radio has to share the air with, and every layer that wraps your data on the way down. Reading a network properly means holding both at once — which is what this course has been for.

Shannon sets the ceiling, modulation decides how close you get to it, and Layer 2 decides how much of that is left for you.

Layer 7 to Layer 1, on one page

Everything this course has covered, in the order a byte meets it

7	**Application**	HTTP, DNS, SMTP — the thing you actually wanted	Class 1
6	**Presentation**	encoding, compression, encryption	Class 1
5	**Session**	opening, holding and closing a conversation	Class 1
4	**Transport**	TCP and UDP: ports, reliability, flow control	Class 1
3	**Network**	IP addresses and routing — getting across networks	Class 1
2	**Data link**	frames, MAC addresses, CRC, who may send — one hop	Classes 1–3
1	**Physical**	voltage, light and radio: line coding, media, modulation	Classes 4–5

The shape of the course: Class 1 walked down from 7 to 3 quickly, then we spent three classes on Layer 2 and two on Layer 1 — because that is where the physics, the arithmetic and most of the exam live.

Seven things to leave the course with

Not facts — the ideas that keep working after you forget the facts

1	**Layers are a contract**	Each layer promises something to the one above and knows nothing about it. That is why the Internet could replace every technology underneath IP without rewriting the web.
2	**Everything shared needs a rule**	One channel and many senders is the same problem whether it is a coaxial cable, a lecture hall or a Teams call. The answers are always partition, contend, or take turns.
3	**Addresses are local or global**	MAC addresses get you across one hop; IP addresses get you across the world. Knowing which is which explains most of what a network does.
4	**Detect, then decide**	A CRC tells you a frame is damaged, never how. What to do about it is a separate decision, made at a different layer.
5	**Physics sets the ceiling**	Shannon caps the channel, attenuation caps the distance, the speed of light caps the delay. No protocol argues with any of them.
6	**Overhead is never zero**	Preambles, headers, acknowledgements, codes and guard intervals all cost. The gap between a specification and a measurement is always one of these.
7	**Ask what problem it solved**	Every odd detail — the 64-byte minimum, the 4B5B table, the 802.11 fourth address — exists because something broke once. Find that, and you never have to memorise it.

THAT IS THE COURSE

From a click in Chiang Mai to a voltage on a wire.

Five classes ago we followed a single request down the stack and asked what each layer had to add. You now know what every one of them does, and more importantly why it had to.

Before you go

Slides for all five classes are on the course page, today included.

Questions

Office hours as usual, and the course chat stays open until the exam.

The exam

Date, coverage and the formula slide are on the course page.

Thank you — it has been a good term.